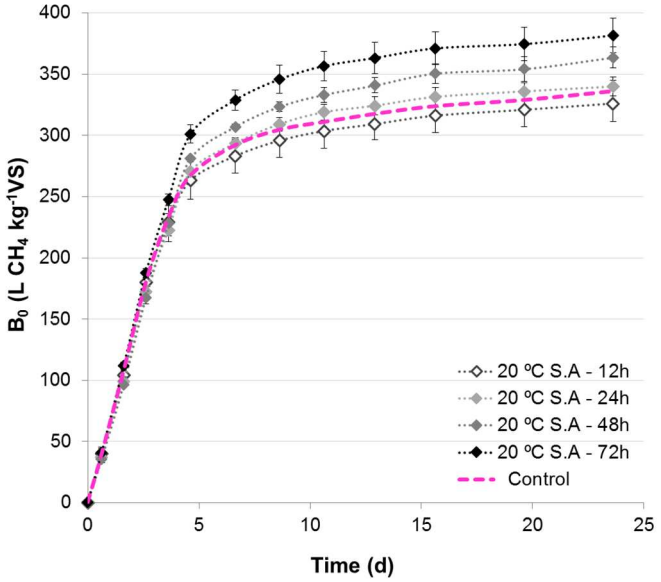
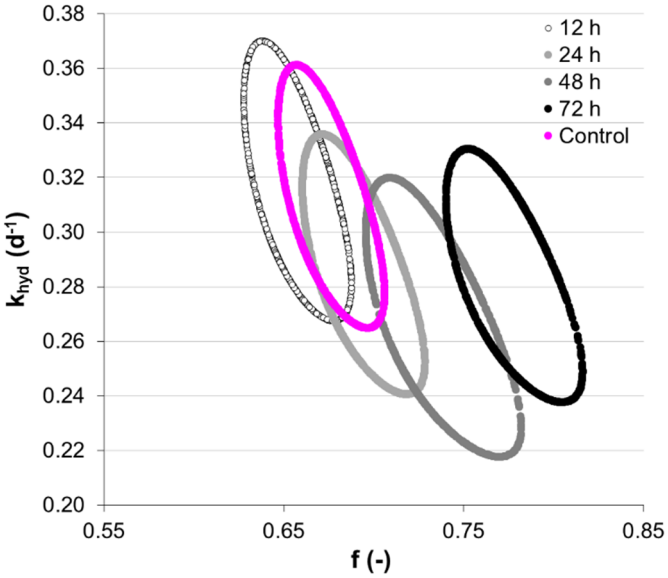
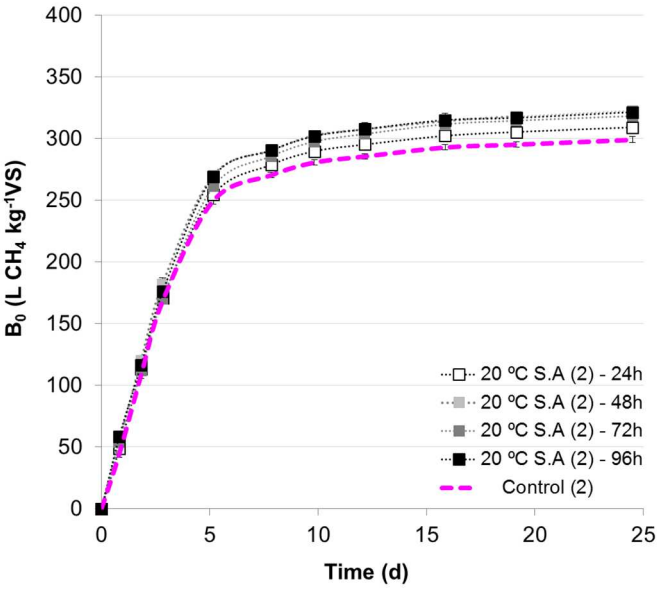
# Cumulative methane production curves and confidence regions



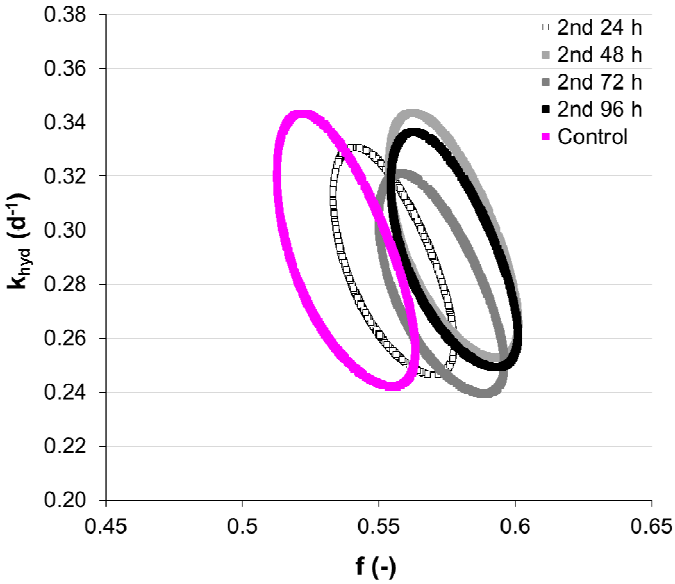

20 ºC Semi-Aerobic (1)



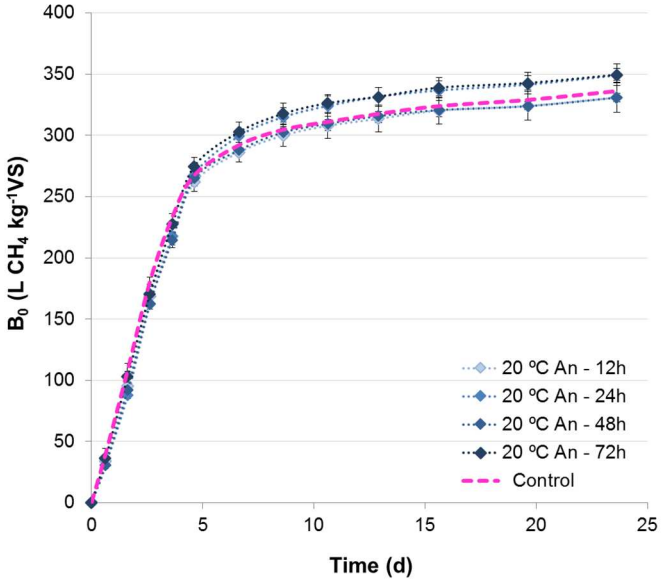

20 ºC Semi-Aerobic (1)



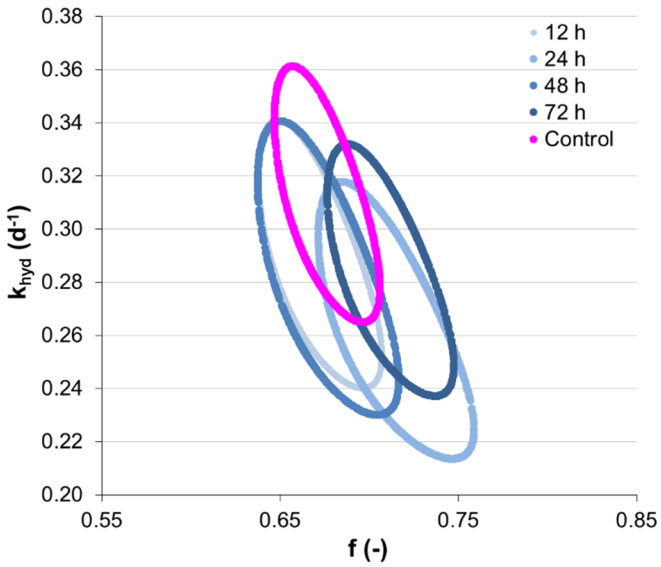

20 ºC Semi-Aerobic (2)



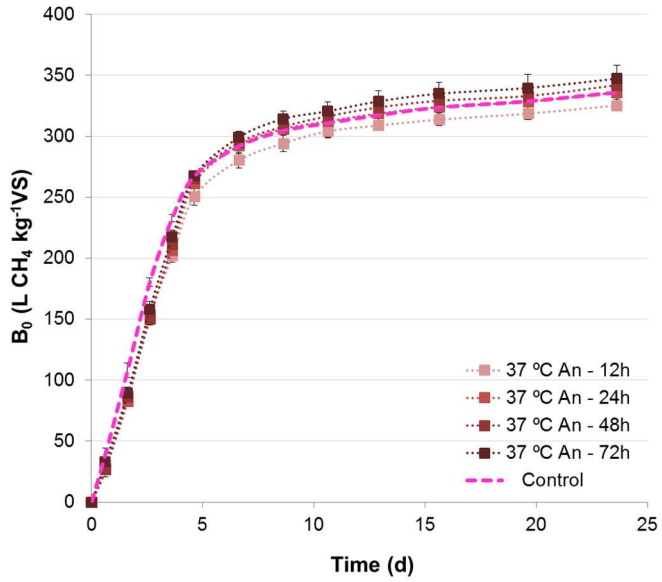

20 ºC Semi-Aerobic (2)



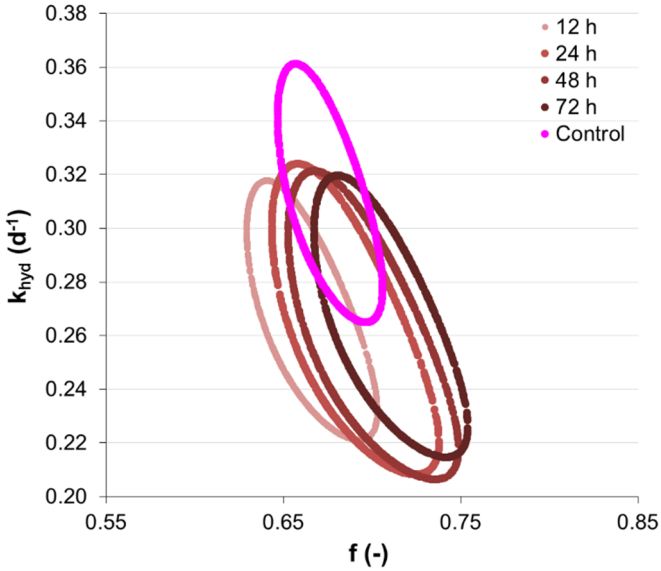

20 ºC Anaerobic




20 ºC Anaerobic




37 ºC Anaerobic




37 ºC Anaerobic

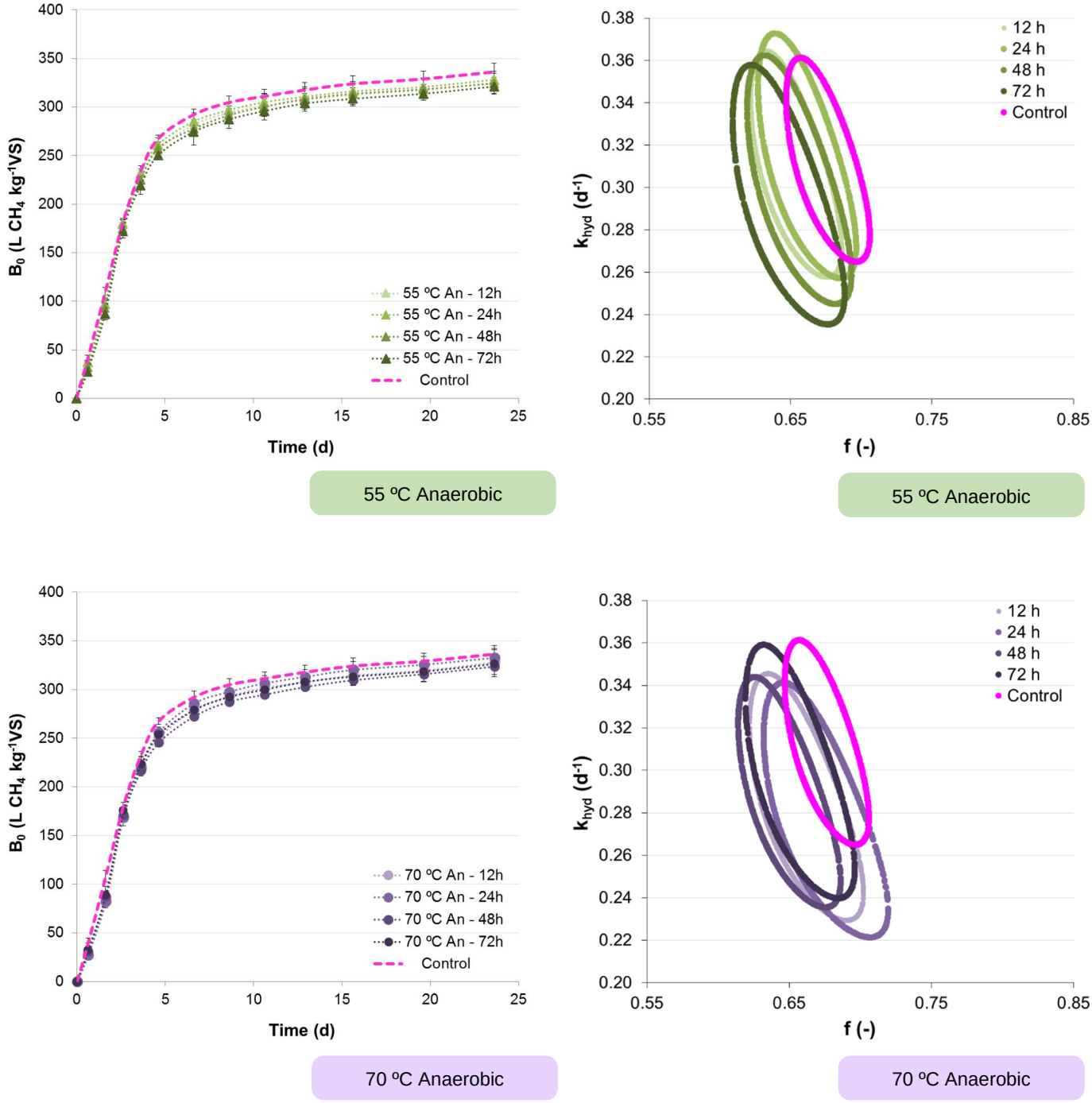



**Fig SI. (Left)** Cumulative specific methane production curves after pre-fermentation at different temperatures, treatment time, and control. **(Right)** Confidence regions from BMP test modelling after pre-fermentation and control. Each ellipse bounds the confidence region (95%) of each trial for the anaerobic biodegradability (f, x-axis) and the hydrolysis rate ($k_{hyd}$, y-axis)

## COD Balance

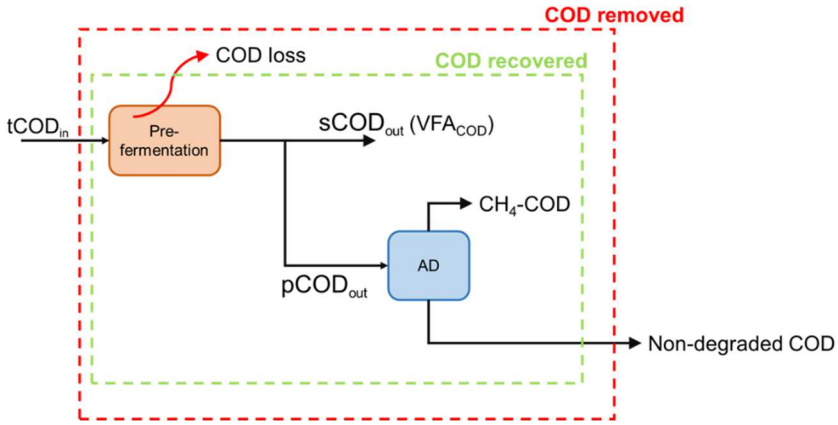



**Table SI.** COD balance for the combined pre-fermentation and anaerobic digestion process

| | tCOD$_{in}$ (g L$^{-1}$) | COD loss (g L$^{-1}$) | pCOD$_{out}$ (g L$^{-1}$) | SCOD$_{out}$ (g L$^{-1}$) | fi (-) | COD recovered (%) | COD lost (%) | COD removed (%) |
|---|---|---|---|---|---|---|---|---|
| **Raw PS (Batch 1)** | | | | | | | | |
| | 57.9 | 0.0 | 55.9 | 2.1 | 0.68 | 69.1 | 0 | 69.1 |
| *20 ºC Semi-aerobic* | | | | | | | | |
| 12 h | 57.9 | 0.6 | 54.6 | 2.7 | 0.66 | 66.8 | 1.1 | 67.9 |
| 24 h | 57.9 | 1.1 | 53.8 | 3.1 | 0.69 | 69.4 | 1.8 | 71.2 |
| 48 h | 57.9 | 3.1 | 50.9 | 3.9 | 0.73 | 70.9 | 5.3 | 76.3 |
| 72 h | 57.9 | 3.0 | 50.9 | 4.1 | 0.78 | 75.5 | 5.1 | 80.7 |
| *20 ºC Anaerobic* | | | | | | | | |
| 12 h | 57.9 | 0.0 | 56.9 | 2.7 | 0.67 | 70.6 | 0.0 | 70.6 |
| 24 h | 57.9 | 2.7 | 52.2 | 3.1 | 0.71 | 69.3 | 4.6 | 73.9 |
| 48 h | 57.9 | 4.6 | 48.9 | 4.4 | 0.68 | 65.0 | 7.9 | 73.0 |
| 72 h | 57.9 | 5.4 | 47.0 | 5.5 | 0.71 | 67.1 | 9.4 | 76.5 |
| *37 ºC Anaerobic* | | | | | | | | |
| 12 h | 57.9 | 0.0 | 54.1 | 4.1 | 0.66 | 68.6 | 0.0 | 68.6 |
| 24 h | 57.9 | 1.7 | 50.6 | 5.6 | 0.69 | 69.9 | 3.0 | 72.9 |
| 48 h | 57.9 | 3.1 | 46.7 | 8.1 | 0.70 | 70.5 | 5.3 | 75.8 |
| 72 h | 57.9 | 8.2 | 40.6 | 9.1 | 0.71 | 65.5 | 14.2 | 79.7 |
| *55 ºC Anaerobic* | | | | | | | | |
| 12 h | 57.9 | 0.0 | 52.2 | 6.7 | 0.65 | 70.2 | 0.0 | 70.2 |
| 24 h | 57.9 | 0.0 | 51.3 | 7.1 | 0.66 | 70.8 | 0.0 | 70.8 |
| 48 h | 57.9 | 0.0 | 49.4 | 8.5 | 0.66 | 71.0 | 0.0 | 71.0 |
| 72 h | 57.9 | 1.2 | 48.2 | 8.5 | 0.65 | 68.8 | 2.0 | 70.9 |
| *70 ºC Anaerobic* | | | | | | | | |
| 12 h | 57.9 | 0.0 | 50.9 | 7.3 | 0.66 | 70.6 | 0.0 | 70.6 |
| 24 h | 57.9 | 0.8 | 49.2 | 7.9 | 0.67 | 70.6 | 1.3 | 72.0 |
| 48 h | 57.9 | 0.0 | 49.1 | 9.4 | 0.65 | 71.3 | 0.0 | 71.3 |
| 72 h | 57.9 | 0.3 | 48.4 | 9.2 | 0.66 | 71.0 | 0.5 | 71.6 |
| **Raw PS (Batch 2)** | | | | | | | | |
| | 72.1 | 0.0 | 69.6 | 2.5 | 0.54 | 55.6 | 0.0 | 55.6 |
| *20 ºC Semi-aerobic* | | | | | | | | |
| 24 h | 72.1 | 1.8 | 66.6 | 3.7 | 0.55 | 55.9 | 2.6 | 58.5 |
| 48 h | 72.1 | 2.7 | 65.5 | 3.9 | 0.58 | 58.1 | 3.7 | 61.9 |
| 72 h | 72.1 | 0.4 | 67.5 | 4.3 | 0.57 | 59.3 | 0.5 | 59.8 |
| 96 h | 72.1 | 0.5 | 66.8 | 4.8 | 0.58 | 60.3 | 0.8 | 61.1 |